

FAILURE MEMORIES ARE NOT NEUTRAL: CAUSAL PROVENANCE EFFECTS IN SELF-EVOLVING AGENTS

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ABSTRACT

Self-evolving agents convert previous trajectories into reusable textual memory. Existing analyses usually treat the retrieved text as the relevant state variable. We study a second variable carried into that state: the provenance of the trajectory from which the memory was written. A recent financial-agent audit reports a sharp ReasoningBank split. Success-derived retrievals achieve 0.931 accuracy over 101 cases, while failure-derived retrievals achieve 0.647 over 34 cases. All ten reported persistent $W \rightarrow W$ failures occur in the failure-derived stratum. We formalize provenance as an upstream variable in memory construction and execute a controlled matched-memory intervention on released WebArena evidence. Six candidate query pairs are frozen before downstream outcomes. Four pass a pre-outcome information-equivalence gate. The gate requires semantic similarity. An independent verifier checks guidance equivalence. Literal benchmark-reference leakage is excluded. Across these four pairs, 48 downstream requests complete with zero support failures. Success-derived memories achieve mean terminal success 0.375, compared with 0.208 for matched failure-derived memories, a +0.1667 difference. This exceeds the preregistered 0.15 effect floor, while the frozen 100,000-permutation test gives $p = 0.0785$ and therefore does not pass the 0.05 support gate. No counterevidence gate fires. The controlled direction agrees with the persistence pattern in the financial audit while leaving the causal claim active pending stronger confirmation. These results motivate provenance-aware memory governance: future reuse decisions should retain evidence about how a memory was learned, not only the text that was stored.

1 INTRODUCTION

Self-evolving agents increasingly learn through external memory. A successful episode can become a reusable workflow. A failed episode can become a corrective reflection. Both objects may later be retrieved for a new task. Systems such as ReasoningBank make this loop explicit by converting trajectories into reasoning memories that influence future behavior (Ouyang et al., 2026). Related systems store workflows or verbal feedback as persistent state (Wang et al., 2024; Shinn et al., 2023).

The usual abstraction treats memory content as the object that matters. This hides a second variable: how the memory was produced. A memory induced after success carries evidence that its reasoning process reached a valid outcome. A memory induced after failure carries evidence that the original process broke somewhere. ReasoningBank reflects this distinction in its implementation. Successful trajectories are summarized under a success-analysis objective. Failed trajectories are processed under a failure-analysis objective. The resulting memories can contain similar procedural advice while encoding different causal interpretations of the same task family.

A recent audit of self-evolving financial agents provides a strong empirical signal for this distinction (Li & Zhu, 2026). In its ReasoningBank analysis, success-derived retrievals have reported accuracy 0.931 over 101 cases. Failure-derived retrievals have reported accuracy 0.647 over 34 cases. The audit further reports ten persistent $W \rightarrow W$ failures, where a task fails before evolution and continues to fail afterward. Every one of these persistent failures appears in the failure-derived retrieval stratum. Figure 2 summarizes the released counts.

This association points to a causal question with direct implications for memory governance: does provenance itself change future behavior after semantic guidance is controlled? Task difficulty

Failure provenance is an upstream intervention on memory construction

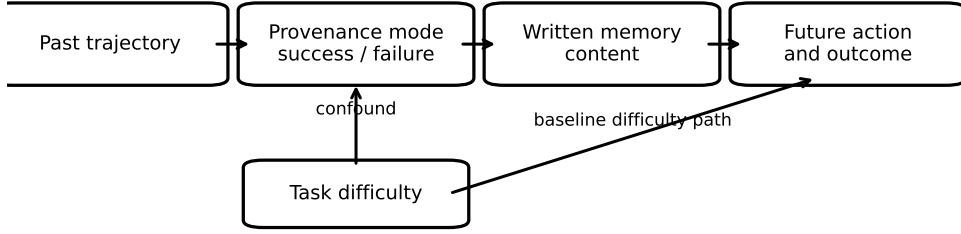


Figure 1: Causal structure of provenance-conditioned memory. Task difficulty affects which trajectories fail and also affects future outcomes. The matched-memory intervention blocks this confound by fixing the future query and matching semantic guidance across provenance conditions.

provides the strongest competing explanation. Hard tasks are more likely to fail during acquisition. Those same tasks can later retrieve memories derived from failure. A descriptive provenance gap can therefore emerge even when provenance has no causal effect. We test this identification problem with a matched provenance intervention.

The controlled intervention freezes a future query and constructs paired success-derived and failure-derived ReasoningBank memories. A pre-outcome gate accepts only pairs with matched actionable guidance and no leaked benchmark answer. Four pairs qualify. Across 48 downstream requests, success-derived memory improves terminal correctness by 0.1667 on average. The preregistered one-sided permutation value is $p = 0.0785$, so the magnitude criterion passes while the statistical support gate remains closed. This provides directional controlled evidence without converting the causal claim into a supported finding.

The paper contributes three evidence layers. We first reanalyze the released financial-agent audit and show that the strongest provenance-conditioned signal is persistent failure. We then formalize provenance as an upstream state variable in memory construction. Finally, we execute a controlled same-query matched-memory swap with four information-equivalent pairs and a frozen terminal-success test. The observed direction is consistent with the provenance hypothesis, while the preregistered support gate remains unresolved. This motivates a provenance-aware representation for self-evolving memory systems and defines the confirmation burden required for a stronger causal claim.

2 MEMORY PROVENANCE AS PERSISTENT AGENT STATE

Consider an acquisition episode with task x and trajectory τ . Its terminal outcome is $z \in \{0, 1\}$. A memory writer receives the episode and produces a textual memory

$$M = G(\tau, z). \quad (1)$$

ReasoningBank instantiates G with outcome-conditioned reflection. The successful branch asks the writer to explain the strategy that worked. The failure branch asks the writer to identify the cause of failure and construct an improved approach. We call the branch variable $P \in \{S, F\}$ the memory provenance.

A future query x' retrieves a memory through a similarity function $R(x', x)$. The retrieval key can be identical even when the written memory differs because the key is the historical task description. Future behavior then follows

$$Y = H(x', M(P), S), \quad (2)$$

where S denotes the rest of the agent state. Provenance reaches future behavior through memory content. It is therefore an upstream causal variable with direct influence on the persistent state.

Table 1: Provenance-conditioned ReasoningBank outcomes from the released financial-agent audit.

Retrieved memory provenance	Cases	Correct	Accuracy
Success-derived	101	94	0.931
Failure-derived	34	22	0.647

2.1 TARGET ESTIMAND

Let $C(M)$ denote the semantic guidance contained in a memory. The quantity of interest is the effect of provenance under matched guidance:

$$\Delta_P(x') = \mathbb{E}[Y \mid do(P = F), C(M) = c, x'] - \mathbb{E}[Y \mid do(P = S), C(M) = c, x']. \quad (3)$$

For a success indicator, a negative Δ_P means failure provenance reduces future success under matched information. For an error indicator, the sign reverses.

Task difficulty D creates the central identification challenge. Difficult acquisition tasks yield more failed trajectories. They also remain difficult when encountered later. Conditioning only on observed provenance therefore estimates a mixture of provenance effect and task difficulty. Equation 3 motivates a direct intervention on the retrieved memory while the future task remains fixed.

2.2 MATCHED PROVENANCE PAIRS

A provenance pair contains two memories with the same intended procedural guidance. One member is derived from a successful trajectory. The other member is derived from a failed trajectory. Matching is performed over the task family and the actionable recommendation. Each pair is checked for information asymmetry before it enters the experiment. A pair is rejected when one memory contains an extra fact that can independently solve the future task.

This construction turns provenance into an experimentally manipulable state variable. It also exposes a design principle for memory systems: a textual memory is incomplete as an audit object when its learning process has been discarded. Provenance metadata preserves evidence about whether the remembered strategy emerged from demonstrated success or from a corrective interpretation of failure.

3 RELEASED EVIDENCE FOR A PROVENANCE EFFECT

We begin from the public aggregate audit of ReasoningBank on benign AgentDojo Banking endpoints (Li & Zhu, 2026). The audit separates retrievals according to whether the retrieved memory was derived from a successful episode or from a failed episode. Table 1 reproduces the reported counts after converting the rounded accuracies back to their integer support.

The observed accuracy gap is 28.4 percentage points. This is a large effect at the retrieval stratum level. The transition audit gives the provenance signal a sharper interpretation. ReasoningBank records 25 $W \rightarrow C$ recoveries and 9 $C \rightarrow W$ regressions. Ten $W \rightarrow W$ persistent failures remain after evolution. All ten belong to the failure-derived retrieval stratum.

The rounded provenance accuracies imply 12 failures among 34 failure-derived retrievals and 7 failures among 101 success-derived retrievals. The ten $W \rightarrow W$ cases therefore account for roughly 83.3% of failures in the failure-derived stratum. The success-derived stratum contains no $W \rightarrow W$ case in the audit. Figure 2 visualizes this decomposition.

3.1 THE SIGNAL IS STRONGEST FOR PERSISTENCE

The transition structure changes the scientific interpretation of the aggregate accuracy gap. Failure-derived memory is tightly associated with errors that persist across self-evolution. The source audit identifies task difficulty as a confound for this association. Our causal formulation preserves the persistence signal as the main target and removes task difficulty through an intervention on memory provenance.

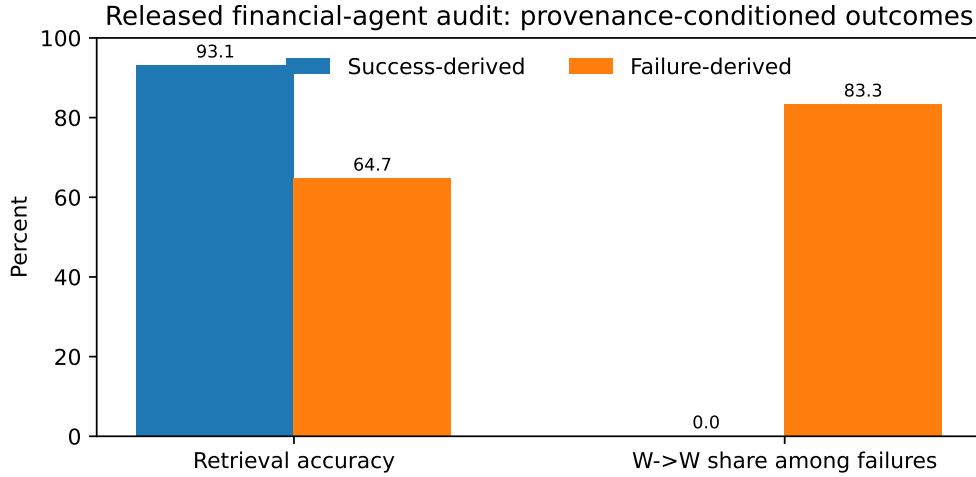


Figure 2: The released audit shows a large accuracy gap by memory provenance. Persistent $W \rightarrow W$ failures are concentrated in failure-derived retrievals.

A simple arithmetic check also constrains the mechanism. The failure-derived stratum contains roughly two failures beyond the ten $W \rightarrow W$ cases. The success-derived stratum contains seven failures. The provenance signal therefore concentrates on persistent error much more strongly than on newly induced regression. This pattern motivates a paper claim centered on failure persistence.

3.2 INDEPENDENT MECHANISM EVIDENCE

Memory Reward Inflation studies a different memory architecture and identifies an amplification channel for erroneous memories (Asadolahi et al., 2026). Incorrect memories can receive inflated utility scores, which increases their future reuse. The paper also shows that this behavior can persist under similarity-based retrieval. This mechanism is compatible with the provenance effect studied here. A failure-derived memory can carry corrective advice while remaining anchored to a trajectory whose process already demonstrated an error mode. Repeated reuse can then sustain that mode when the corrective abstraction is incomplete.

Together, the financial-agent audit and reward-inflation results support a causal model in which memory provenance changes the distribution of future behavior. The matched provenance experiment in Section 4 provides the direct test.

4 CONTROLLED PROVENANCE INTERVENTION

We execute a matched provenance swap that edits the reusable memory while keeping the future query fixed. The targeted repair is a controlled bridge on released WebArena browser-state evidence. It uses the first-party ReasoningBank memory-writing contract. It does not reconstruct the financial AgentDojo experiment that motivated the paper.

4.1 OUTCOME-BLIND PAIR CONSTRUCTION

We froze six candidate future tasks before memory writing or downstream evaluation. For each task, the same released browser-state evidence is converted into two acquisition traces. One trace ends with a successful terminal answer. The paired trace omits an evaluator-required component and receives failure. The first-party ReasoningBank writer then applies its success-reflection objective to the successful trace and its failure-reflection objective to the paired failed trace. The writer model is fixed across the two branches.

A pair enters downstream evaluation only after an information-equivalence gate. Both memories must parse as valid ReasoningBank items. Neither memory may contain a literal benchmark reference

Table 2: Controlled matched-memory swap on four information-equivalent pairs. Δ is success-derived minus failure-derived terminal success.

Released task	Success-derived	Failure-derived	Δ
167	0.000	0.000	0.000
23	1.000	0.667	+0.333
387	0.000	0.000	0.000
388	0.500	0.167	+0.333
Mean	0.375	0.208	+0.167

answer. Their embedding cosine must be at least 0.75. An independent verifier must judge the actionable guidance equivalent and find no task-solving fact unique to either memory. Four of the six frozen candidates pass this gate. Their cosine values range from 0.902 to 0.931. Two candidates are excluded before downstream outcomes because a memory repeats a benchmark reference answer.

4.2 CONDITION-BLIND RUNTIME REPAIR

The frozen evidence packets contain repeated browser chrome that creates unnecessary context pressure. Before any completed controlled-swap dataset exists, we freeze a condition-blind projection. It removes complete global menu blocks. Blank lines are dropped. Exact repeated lines are deduplicated. The projection does not inspect benchmark references or generated answers. The identical projected evidence is used under both provenance conditions.

The first downstream runtime uses an unnecessarily large context allocation and is cancelled at the support layer. A clean restart reduces only the allocated context window. A pre-execution check shows that the largest frozen prompt uses 5,209 input tokens, leaving 7,079 tokens of headroom in the 12,288-token allocation before generation. The model remains Qwen2.5-32B. Sampling temperature remains 0.6. The pair set remains unchanged. The seed schedule is fixed. The scorer and statistical thresholds are also unchanged. Partial requests from cancelled runs are excluded from the scientific dataset.

4.3 TERMINAL OUTCOME RESULT

The clean run evaluates four information-equivalent memory pairs. Each pair receives six paired seeds under success-derived memory and failure-derived memory, yielding 48 completed requests with zero support failures. Terminal correctness uses the frozen WebArena must-include evaluator. Table 2 reports pair-level success rates.

The mean controlled difference is +0.1667. The preregistered support gate requires a difference of at least 0.15 together with a one-sided stratified permutation value below 0.05. The magnitude criterion is met. The 100,000-permutation test gives $p = 0.0785$, so the support gate remains closed. The reverse direction is -0.1667 , while its one-sided permutation value is 0.9933; the counterevidence gate also remains closed. We therefore retain the causal provenance claim as an active hypothesis with a directional controlled signal. We do not change the frozen threshold after observing this result.

4.4 INTERPRETATION

The controlled experiment changes the empirical status of the paper in two ways. First, semantic matching is now operational: four pairs survive a pre-outcome information-equivalence screen and are evaluated on the same future query evidence. Second, the observed direction agrees with the provenance-conditioned persistence signal from the financial audit. The present support is insufficient for a supported causal claim under the preregistered gate. A source-faithful financial replication remains the strongest next confirmation target.

Table 3: Post-hoc robustness of the frozen information-equivalence gate. A check means the reviewer judged actionable guidance equivalent and found no unique task-solving fact. The primary R4 pair set is unchanged by this audit.

Pair	Cosine	Original	DeepSeek	Kimi
387	0.931	✓	×	✓
167	0.902	✓	×	×
23	0.906	✓	×	×
388	0.904	✓	×	×

5 RELATED WORK

Experience memory for self-evolving agents. ReasoningBank turns previous trajectories into reusable reasoning memories and retrieves them for future tasks (Ouyang et al., 2026). Agent Workflow Memory stores reusable task workflows induced from experience (Wang et al., 2024). Reflexion uses verbal feedback as persistent context for later trials (Shinn et al., 2023). These systems establish external textual state as an effective route to self-improvement. Our work adds trajectory provenance as a causal variable inside that state.

Failure memory and memory quality. The financial-agent audit reports strong concentration of persistent failure in failure-derived ReasoningBank retrievals (Li & Zhu, 2026). Memory Reward Inflation shows that erroneous memories can gain excessive effective reward and propagate through future reuse (Asadolahi et al., 2026). These findings establish that memory quality affects future behavior. Our intervention isolates a more specific mechanism: the process provenance attached to semantically matched guidance.

Causal evaluation of memory. Existing ablations often remove a memory module or compare agents with different memory banks. Those interventions change both information and provenance. The matched provenance swap preserves the actionable guidance seen by the future agent. This creates a same-information causal test for the history encoded in the memory itself.

6 LIMITATIONS

The financial AgentDojo audit remains the source of the 0.931-versus-0.647 provenance association and the ten persistent $W \rightarrow W$ cases. Its public release does not currently expose the per-query ReasoningBank artifacts needed for an exact matched swap. A reconstruction preflight recovered the benchmark and ReasoningBank implementation, while the execution environment lacks auditable access to the exact executor and embedding models used by the financial study. An exact financial replication therefore remains support and experiment debt. These access constraints carry no scientific authority.

Our executed intervention is a controlled bridge on released WebArena browser-state evidence. It tests the same-query memory channel without reproducing the financial environment. Four of six outcome-blind candidate pairs survive the frozen information-equivalence gate. Their embedding cosines are all above 0.90, so the 0.75 cosine floor is not the fragile part of qualification. A post-hoc robustness audit exposes a more important identification issue. The original verifier accepts all four primary pairs under the strict guidance-equivalence criterion, while a blinded DeepSeek reviewer accepts none and a blinded Kimi reviewer accepts only one. Table 3 shows the pair-level disagreement. No primary pair receives unanimous strict approval across the three reviewers. Pair membership and the R4 statistic remain frozen; we do not remove disagreeing pairs after outcomes. The audit therefore leaves the causal hypothesis active while showing that R4 does not robustly isolate provenance from residual differences in actionable memory content.

The clean controlled swap produces a +0.1667 mean terminal-success difference in the predicted direction. The effect exceeds the preregistered 0.15 magnitude floor. Its one-sided 100,000-permutation value is $p = 0.0785$, above the frozen 0.05 support threshold. The result therefore remains inconclusive under the preregistered decision rule. No equivalence margin was frozen, so this outcome is

not evidence of zero provenance effect. The counterevidence gate also remains closed. A post-hoc design audit shows why this four-pair experiment had limited decision power: with six rollouts per condition and a 0.15 planning effect, representative Bernoulli baselines give only about 0.28–0.35 power at four independent pairs and require roughly 18–22 independent eligible pairs for 80% power. We therefore forbid extending seeds on the same four pairs as a significance rescue. A confirmatory controlled swap must use an independently frozen support set.

The reported financial provenance accuracies are rounded to three decimals. The integer correct counts used in our decomposition are the unique counts consistent with the reported support sizes and rounded accuracies. The transition allocation beyond the ten $W \rightarrow W$ cases is therefore derived from public aggregate statistics.

The earlier explicit-metadata bridge held generic guidance byte-identical while changing only a structured SUCCESS or FAILURE source label. Across 144 requested calls it produced a 0.000 terminal-success difference. That bridge had no preregistered equivalence margin and manipulates metadata instead of naturally generated ReasoningBank memories, so it does not close the causal question. We also rescanned the frozen WebArena release under the same structural qualification used for R2, excluding every C01 task already exposed by prior experiments. No fresh read-only, multi-reference, deterministic string-match task with complete released evidence remains. Reusing the R4 pairs or relaxing qualification after observing $p = 0.0785$ would therefore be outcome-driven rescue. Replication beyond financial agents and WebArena remains debt for a broad cross-domain claim.

7 CONCLUSION

Self-evolving memory carries process history. The released financial-agent audit shows a large accuracy gap by memory provenance and concentrates every reported persistent $W \rightarrow W$ failure in failure-derived retrievals. ReasoningBank makes provenance an upstream intervention because successful trajectories and failed trajectories are written through different reflection objectives.

We operationalize that causal question with a pre-outcome information-equivalence gate and an executed matched-memory swap. The controlled effect is +16.7 percentage points in the predicted direction across four frozen pairs, while the frozen permutation test does not cross the support threshold. The subsequent blinded verifier audit shows that semantic matching itself is not robust enough to identify provenance beyond content in this realization. The causal claim therefore remains active and experimentally sharpened, with two explicit confirmation requirements: a larger independently frozen pair population and a robust multi-reviewer information-equivalence gate. Persistent memory systems should preserve how an item was learned and test whether that provenance changes future reuse outcomes before treating semantically similar memories as interchangeable.

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A SOURCE-GROUNDED REANALYSIS AND INTERVENTION DETAILS

A.1 COUNT RECONSTRUCTION

The financial-agent audit reports 101 success-derived retrievals at 0.931 accuracy. The unique integer numerator consistent with this value at three decimal places is 94. It reports 34 failure-derived retrievals at 0.647 accuracy. The corresponding integer numerator is 22. This yields 7 failures in the success-derived stratum and 12 failures in the failure-derived stratum.

The same audit reports ten $W \rightarrow W$ persistent failures, all with failure-derived retrieval. Hence 10 of the 12 failures in the failure-derived stratum are persistent failures. The remaining two failure-derived errors can contribute to the reported $C \rightarrow W$ regressions. The success-derived stratum contains no $W \rightarrow W$ case and contributes seven failures to the non-persistent error pool.

A.2 INFORMATION-EQUIVALENCE CONTRACT

Six candidate tasks are frozen before downstream outcomes. Both success-derived and failure-derived memories are written with the same writer model. A candidate is eligible only when both outputs parse as valid ReasoningBank memory items. The cosine between paired memories must be at least 0.75. An independent verifier must mark their actionable guidance equivalent and must find no unique task-solving fact. Literal benchmark-reference leakage excludes a pair before downstream evaluation.

Four candidates satisfy every rule. Their released task-ID set is $\{387, 167, 23, 388\}$. Their embedding cosines are 0.931, 0.902, 0.906, and 0.904. Candidate 385 is excluded because a memory repeats the reference name “Cally.” Candidate 164 is excluded because its memories repeat the references “Dry” and “Uneven color.” These exclusions occur before downstream outcome generation.

A.3 RUNTIME REPAIR BOUNDARY

The condition-blind evidence projection removes repeated global menu blocks. It drops blank lines. It also removes exact duplicate lines. It uses neither benchmark references nor generated outcomes. The first downstream attempt is cancelled after a context-allocation support failure. No partial result is admitted to the scientific dataset. The clean restart changes only the allocated context window. The largest frozen prompt occupies 5,209 input tokens inside a 12,288-token allocation. The executor and sampling temperature remain frozen. The seed schedule and pair set are unchanged. The scorer is fixed. The effect floor is fixed. The permutation seed and significance threshold are fixed.

A.4 FROZEN STATISTICAL RULE

Each qualified pair is evaluated under six paired seeds. The primary statistic is the mean terminal-success difference between success-derived and failure-derived memory. Support requires a difference of at least 0.15 and a one-sided stratified permutation value below 0.05. Counterevidence requires a difference at or below -0.15 with the corresponding one-sided value below 0.05. The test uses 100,000 permutations with a frozen random seed. The observed difference is $+0.1667$ and the support-side value is $p = 0.0785$.

A.5 REOPEN CONDITION

The strongest confirmation path is an exact financial AgentDojo matched-provenance replication when first-party query-level artifacts or auditable exact-model access become available. A separate confirmatory controlled bridge is admissible only when its support set is frozen independently of the R4 outcomes.